R Course
Assignment 1

Introduction

In this assignment, you will use basic R programming to simulate how insecticide resistance evolves in an insect population.

Consider a single gene with two alleles: S (susceptible to the insecticide, dominant) and r (resistant, recessive). Because r is recessive, only insects with genotype rr clearly survive spraying, insects with genotype SS or Sr still behave as susceptible. We describe this with a survival probability between 0 and 1 for each genotype, for example:

```
w_SS <- 0.35 # the probability of surviving to reproduce of genotype SS
w_Sr <- 0.35 # the probability of surviving to reproduce of genotype Sr
w_rr <- 1.0  # the probability of surviving to reproduce of genotype rr
```

Let q_n denote the relative frequency of the resistance allele r in generation n, and let $p_n = 1 - q_n$ be the relative frequency of S. Assuming random mating, the genotypes SS, Sr and rr occur in generation n with the Hardy-Weinberg proportions p_n^2, $2p_nq_n$ and q_n^2. After one round of selection by the insecticide, the frequency of the resistance allele in generation $n+1$ is

$$q_{n+1} = \frac{p_n q_n w_{Sr} + q_n^2 w_{rr}}{\bar{w}_n}, \qquad \bar{w}_n = p_n^2 w_{SS} + 2p_n q_n w_{Sr} + q_n^2 w_{rr},$$

where $\bar{w}_n$ is the mean survival probability in generation n.

Task

Create a new R script called `Assignment1_solution.R`.

1. Create a variable for the number of generations to simulate (25).

```
number_of_generations <- 25
```

2. Create variables `w_SS`, `w_Sr` and `w_rr` for the three genotype survival probabilities given above.

(As above)

3. Create a vector `generation` containing the numbers 0 to `number_of_generations`.

```
generations <- 0:number_of_generations
generations
```

```
##  [1]  0  1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24
## [26] 25
```

4. Create a numeric vector `q` of length `number_of_generations + 1`, and set `q[1] <- 0.05` as the starting frequency of the resistance allele, before spraying began.

```
q <- rep(0, length.out = number_of_generations + 1)
q[1] <- 0.05
q
```

```
##  [1] 0.05 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
## [16] 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
```

5. Use a `for` loop to fill in the remaining entries of `q` based on the formulas above.

```
for ( n in 1:(length(q)-1) ) {
  q_n <- q[n]
  p_n <- 1 - q_n
  w_bar <- p_n * p_n * w_SS + 2 * p_n * q_n * w_Sr + q_n * q_n * w_rr
  q[n+1] <- (p_n * q_n * w_Sr + q_n * q_n * w_rr) / w_bar
}
q
```

```
##  [1] 0.05000000 0.05439033 0.05955712 0.06571164 0.07314426 0.08226277
##  [7] 0.09365337 0.10818014 0.12715069 0.15259403 0.18771983 0.23761301
## [13] 0.30996596 0.41444741 0.55606163 0.71799776 0.85592988 0.93896814
## [19] 0.97685883 0.99165237 0.99704641 0.99896227 0.99963630 0.99987265
## [25] 0.99995542 0.99998440
```

6. Modify the loop to also save the mean survival probability $\bar{w}_n$ in a vector `w_bar` of length `number_of_generations`.

```
w_bar <- rep(0, length.out = number_of_generations)
for ( n in 1:(length(q)-1) ) {
  q_n <- q[n]
  p_n <- 1 - q_n
  w_bar[n] <- p_n * p_n * w_SS + 2 * p_n * q_n * w_Sr + q_n * q_n * w_rr
  q[n+1] <- (p_n * q_n * w_Sr + q_n * q_n * w_rr) / w_bar[n]
}
q
```

```
##  [1] 0.05000000 0.05439033 0.05955712 0.06571164 0.07314426 0.08226277
##  [7] 0.09365337 0.10818014 0.12715069 0.15259403 0.18771983 0.23761301
## [13] 0.30996596 0.41444741 0.55606163 0.71799776 0.85592988 0.93896814
## [19] 0.97685883 0.99165237 0.99704641 0.99896227 0.99963630 0.99987265
## [25] 0.99995542 0.99998440
```

```
w_bar
```

```
##  [1] 0.3516250 0.3519229 0.3523056 0.3528067 0.3534776 0.3543987 0.3557011
##  [8] 0.3576069 0.3605087 0.3651352 0.3729052 0.3866990 0.4124513 0.4616483
## [15] 0.5509830 0.6850885 0.8262004 0.9230798 0.9702646 0.9891934 0.9961660
## [22] 0.9986516 0.9995273 0.9998344 0.9999420
```

7. To see how the strength of the insecticide affects the speed of resistance evolution, repeat the simulation for a weaker insecticide (`w_SS <- 0.6` and `w_Sr <- 0.6`). Use a `for` loop, as before, to compute this second frequency trajectory and store it in a new vector `q_weak`.

```
w_SS_weak <- 0.6
w_Sr_weak <- 0.6
q_weak <- rep(0, length.out = number_of_generations + 1)
q_weak[1] <- 0.05
w_bar_weak <- rep(0, length.out = number_of_generations)
for ( n in 1:(length(q)-1) ) {
  q_n <- q_weak[n]
  p_n <- 1 - q_n
  w_bar_weak[n] <- p_n * p_n * w_SS_weak + 2 * p_n * q_n * w_Sr_weak + q_n * q_n * w_rr
  q_weak[n+1] <- (p_n * q_n * w_Sr_weak + q_n * q_n * w_rr) / w_bar_weak[n]
}
q_weak
```

```
##  [1] 0.05000000 0.05158070 0.05325994 0.05504693 0.05695199 0.05898679
##  [7] 0.06116454 0.06350023 0.06601097 0.06871631 0.07163875 0.07480422
## [13] 0.07824279 0.08198946 0.08608518 0.09057812 0.09552524 0.10099423
## [19] 0.10706608 0.11383822 0.12142857 0.12998080 0.13967098 0.15071619
## [25] 0.16338552 0.17801404
```

```
w_bar_weak
```

```
##  [1] 0.6010000 0.6010642 0.6011346 0.6012121 0.6012974 0.6013918 0.6014964
##  [8] 0.6016129 0.6017430 0.6018888 0.6020528 0.6022383 0.6024488 0.6026889
## [15] 0.6029643 0.6032818 0.6036500 0.6040799 0.6045853 0.6051837 0.6058980
## [22] 0.6067580 0.6078032 0.6090861 0.6106779
```

8. To see how the starting frequency of the resistance allele interacts with the strength of selection, repeat the simulation once more using a lower starting frequency `q[1] <- 0.01` combined with a much stronger insecticide (`w_SS <- 0.1` and `w_Sr <- 0.1`). Use a `for` loop, as before, to compute this trajectory and store it in a new vector `q_lowstart`. Even though it starts out behind the other two scenarios, the stronger selection should let it overtake both of them at some point during the simulation.

```
w_SS_lowstart <- 0.1
w_Sr_lowstart <- 0.1
q_lowstart <- rep(0, length.out = number_of_generations + 1)
q_lowstart[1] <- 0.01
w_bar_lowstart <- rep(0, length.out = number_of_generations)
for ( n in 1:(length(q)-1) ) {
  q_n <- q_lowstart[n]
  p_n <- 1 - q_n
  w_bar_lowstart[n] <- p_n * p_n * w_SS_lowstart + 2 * p_n * q_n * w_Sr_lowstart + q_n * q_n * w_rr
  q_lowstart[n+1] <- (p_n * q_n * w_Sr_lowstart + q_n * q_n * w_rr) / w_bar_lowstart[n]
}
q_lowstart
```

```
##  [1] 0.01000000 0.01089020 0.01194482 0.01321196 0.01475977 0.01668771
##  [7] 0.01914605 0.02237138 0.02675517 0.03298522 0.04236262 0.05758388
## [13] 0.08489353 0.14063391 0.27048785 0.56013024 0.88496266 0.98570686
## [19] 0.99853322 0.99985293 0.99998529 0.99999853 0.99999985 0.99999999
## [25] 1.00000000 1.00000000
```

```
w_bar_lowstart
```

```
##  [1] 0.1000900 0.1001067 0.1001284 0.1001571 0.1001961 0.1002506 0.1003299
##  [8] 0.1004504 0.1006443 0.1009792 0.1016151 0.1029843 0.1064862 0.1178001
## [15] 0.1658473 0.3823713 0.8048430 0.9744562 0.9973617 0.9997353 0.9999735
## [22] 0.9999974 0.9999997 1.0000000 1.0000000
```

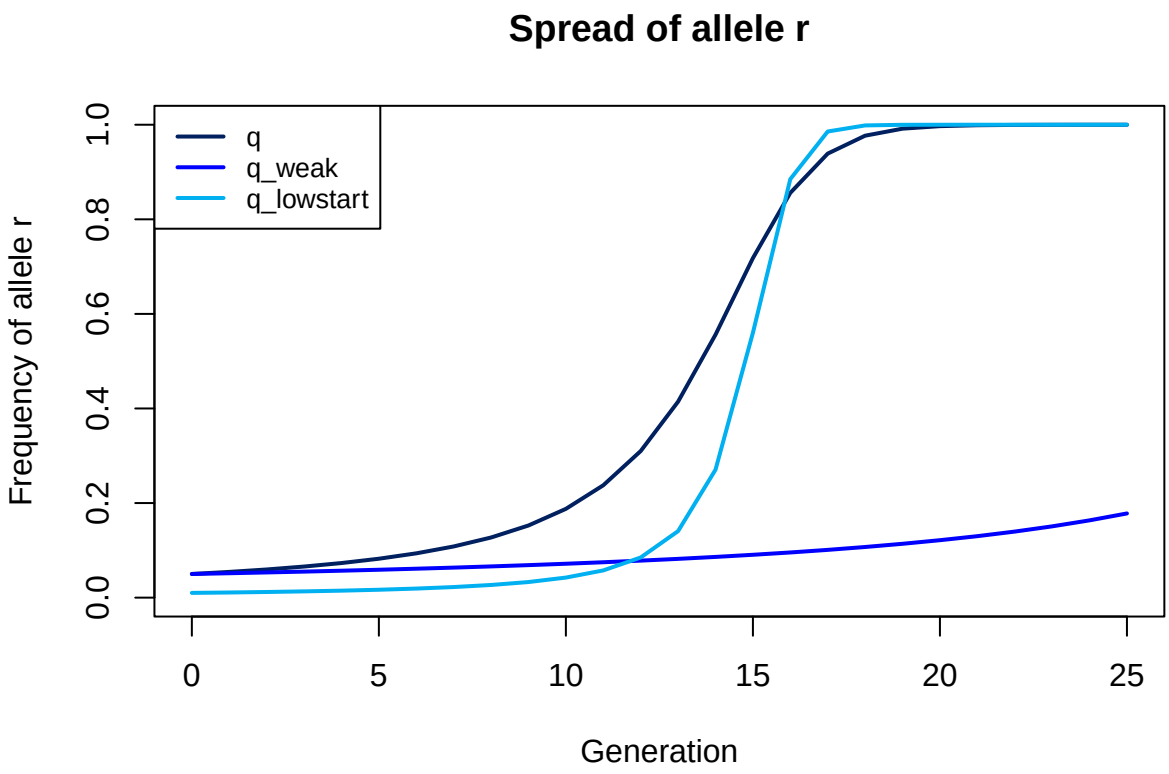
9. Use a `while` loop to find the first generation in which `q_lowstart` overtakes `q`, and print that generation.

```
index <- 1
while (q_lowstart[index] <= q[index]) {
  index <- index + 1
}
paste("Generation:", index)
```

```
## [1] "Generation: 17"
```

10. Create a plot of `q`, `q_weak` and `q_lowstart` against `generation`. Add a title, labels for both axes and a legend distinguishing the three scenarios.

```
plot(generations, q, type = "l", col = "#002060", lwd = 2, ylim = c(0, 1),
     main = "Spread of allele r",
     xlab = "Generation", ylab = "Frequency of allele r")
lines(generations, q_weak, col = "blue", lwd = 2)
lines(generations, q_lowstart, col = "#00B0F0", lwd = 2)
legend(
  "topleft",
  legend = c("q", "q_weak", "q_lowstart"),
  col = c("#002060", "blue", "#00B0F0"), lwd = 2, cex = 0.8, bg = "white"
)
```



11. Add a short comment explaining what the simulation suggests about the effect of insecticide strength and starting frequency on how quickly resistance spreads.

The simulation suggests that the starting frequency does not matter for the spread of the allele. The strength of the insecticide determines the rate at wich the resistance spreads.